FG-GAN: A Fine-Grained Generative Adversarial Network for Unsupervised SAR-to-Optical Image Translation

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***Abstract*— Synthetic aperture radar (SAR) and optical sensing are two important means of Earth observation. SAR can be used for all-day and all-weather Earth observation, but it has the disadvantages of speckle noise and geometric distortion, which are not conducive to human eye recognition. Optical image conforms to the characteristics of human visual observation, but it is easily affected by climate and time. Therefore, to integrate the advantages of the two, researchers have carried out extensive work on SAR-to-optical (S2O) image translation. Most of the existing methods for S2O image translation are supervised and need paired training samples, limiting its large-scale application in remote sensing field. Thus, we give priority to an unsupervised S2O image translation method. Meanwhile, we find that the images generated by unsupervised methods suffer from significant detail deficiencies. To solve this problem, we propose a fine-grained generative adversarial network (FG-GAN) introducing three strategies to enhance the detailed information in generated optical images. First, we design an unbalanced generator (UBG) with complex encoder networks and relatively simple decoder networks. The complex encoder extracts abundant feature information, while the decoder obtains key details by filtering these features. Second, to match the learning ability of the generator, we present a multiscale discriminator (MSD) to enhance the discriminant ability of the network. Third, we propose a comprehensive normalization group (CNG) to promote the physical representation consistency of SAR and optical images. Extensive experiments have been conducted, and the results show that our method is superior to the state-of-the-art (SOTA) methods on both subjective and objective evaluation indicators. Moreover, our FG-GAN has a significant improvement on classification accuracy, indicating its potential in facilitating the performance of practical remote sensing tasks.**

***Index Terms*— Comprehensive normalization group (CNG), multiscale discriminator (MSD), synthetic aperture radar (SAR)-to-optical (S2O) image translation, unbalanced generator (UBG).**

I. Introduction

With the development of remote sensing technology, more and more synthetic aperture radar (SAR) and optical sensing images are applied to mapping, navigation, and target detection. SAR is a kind of microwave imaging radar which can produce high-resolution images. But the color information of land cover type presented by SAR images is very different from that of optical images, and there are geometric distortions and shadows in SAR images, which seriously affect the visual effect of SAR imaging. It is difficult for nonremote sensing professionals to intuitively identify land cover types and obtain useful information from SAR images, resulting in a waste of resources. In fact, optical images are consistent with human eye observation and their susceptibility to cloud can be exactly compensated by SAR images. Therefore, how to better combine SAR with optical images to make them more suitable for human visual system and more convenient for future application has become a research hotspot in the field of remote sensing. Due to the rapid development of image translation technology, we consider using image-to-image translation methods to translate SAR images into optical images. In recent years, many image-to-image translation methods based on deep learning have been proposed. Among them, the methods based on generative adversarial network (GAN) [1] have achieved great results. The emergence of GANs provides a new idea for SAR-to-optical (S2O) image translation, which can alleviate the problem of SAR visual observation.

However, most of the existing methods for S2O image translation are supervised and need paired training samples, such as conditional generative adversarial nets (CGAN) [2], Pix2pix [3], and Pix2pixHD [4]. In fact, paired remote sensing image datasets are not readily available, and it is a difficult problem to register SAR and optical images in pixels because of the large differences in imaging principles, which seriously hinders the development of S2O image translation and affects the application value of SAR images.

Unsupervised image-to-image translation models such as CycleGAN [5], MUNIT [6], UGATIT [7], and DivCoDRIT [8] do not require paired training images, which reduce the difficulty of data registration. However, there are several problems with the existing unsupervised image-to-image translation methods. First, the generator structure using the residual network is relatively simple, resulting in imperfect feature extraction and weak correlation between features. Second, the ability of the discriminator to distinguish detailed information of the image is insufficient. Third, the physical representation information of SAR and optical images is not taken into account. As a result, there are obvious detail missing and color

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distortion in the generated images, which is not conducive for nonremote sensing professionals to correctly identify the land type.

In this article, we give priority to an unsupervised S2O image translation method. We use CycleGAN as a baseline model and find that the optical images generated by CycleGAN are not fine enough. Therefore, we propose a fine-grained GAN (FG-GAN) for unsupervised S2O image translation. The fine-grained effects are achieved via three main aspects.

A. Unbalanced Generator

We apply dense connection modules and residual modules in the generator to form an unbalanced network structure. Each layer of the dense connection module contains the feature information of all previous layers. It guarantees maximum flow of information between networks and alleviates gradient disappearance. Therefore, with the deepening of the network, more shallow feature information is retained, which increases the richness of feature extraction. The residual module is used to filter these features and improve the attention of the generator to some detailed textures.

B. Multiscale Discriminator

Compared with the network structure of unbalanced generator (UBG), the ability of a single discriminator is not enough to match the generator, affecting the adversarial training of the two. Therefore, we present a multiscale discriminant framework, which comprehensively used both local and global perspectives to constrain the details and styles of generated images. The discriminator pays more attention to the detailed differences between the generated and real images, thus encouraging the generator to produce more realistic images.

C. Comprehensive Normalization Group

To further generate fine-grained optical images, and considering that the different physical representations of SAR and optical images affect the quality of image translation, we propose comprehensive normalization group (CNG) to enhance the consistency between the generated and optical images. It includes batch normalization (BN), instance normalization (IN), and layer normalization (LN) for restricting the physical representation information of SAR and optical images, so as to improve the quality and fidelity of the generated images.

To sum up, FG-GAN is an unsupervised S2O image translation method that can generate fine-grained images. After a large number of experiments, FG-GAN is superior to the state-of-the-art (SOTA) methods and improves the utilization of SAR images in practical remote sensing tasks.

II. Related Work

A. Generative Adversarial Network

Goodfellow *et al.* [1] proposed GAN in 2014 and defined its framework structure and objective function. It pointed out that GAN consists of two types of networks, generator and discriminator. The generator is trained to learn the distribution of real data, while the discriminator is used to distinguish between the generated and real input data. Moreover, GAN can transform the input noise into an image and improve the quality of the generated image using adversarial training. However, there exists gradient disappearance in the training process of GAN. To alleviate the problem, least squares generative adversarial fetworks (LSGAN) [9] adopted the loss function based on the least-square method, and Wasserstein GAN (WGAN) [10] used Earth mover's distance to replace Jensen–Shannon divergence which better measured the distance between the generated and real data.

GANs also have great development in network structure. deep convolutional generative adversarial networks (DCGAN) [11] first proposed the idea of combining CNN with GAN to improve the quality of generated images and the convergence speed of the model. However, due to the limitation of the local receptive field in convolution network, it was unable to generate a wide range of relevant regions. Therefore, self-attention generative adversarial networks (SAGAN) [12] proposed to use the self-attention mechanism network to eliminate the influence of receptive field in convolution and improve the integrity of generated images. Then, to get more precise and realistic images, BigGAN [13] was proposed and got higher inception scores [14]. Recently, Wu *et al.* [15] presented a novel gradient normalization method for stabilizing the training of GANs.

With the increasing development of GANs, numerous related derivative models have emerged, making the application scope of GANs larger and larger, especially in the field of computer vision.

1) Image Generation: CGAN was proposed in 2014 to make the generated images as real as possible and still meet the requirements of semantic conditions. Later, people began to pay attention to the fidelity and clarity of the generated images. For example, StackGAN [16] and PGGAN [17] could generate large clear images with high resolution. Subsequently, Karras *et al.* presented StyleGAN [18] on the basis of PGGAN. The model mapped the potential encoding to the hidden space through the mapping network. Then, it used the intermediate hidden variables to realize the decoupling of the mapping relationship, which further improved the resolution of generated images. In addition, for the task of large shift between the spectral distribution of the training and test data, ColorMapGAN [19] was proposed to transform the colors of the training data to test data. Ling *et al.* [20] introduced a new method for high-quality, high-precision semantic image editing based on GAN, allowing users to edit images by modifying some details.

2) Style Transfer: Pix2pix established a corresponding relationship between the output and input that realized image-to-image translation. In addition, unsupervised style transfer has also developed greatly. CycleGAN used a cycle-consistent loss function to achieve unsupervised image translation. UGATIT added an attention mechanism and a learnable normalization function to both the generator and the discriminator, which presented a new unsupervised image translation strategy and improved the style transfer ability of the model. To obtain a more compact and efficient architecture by

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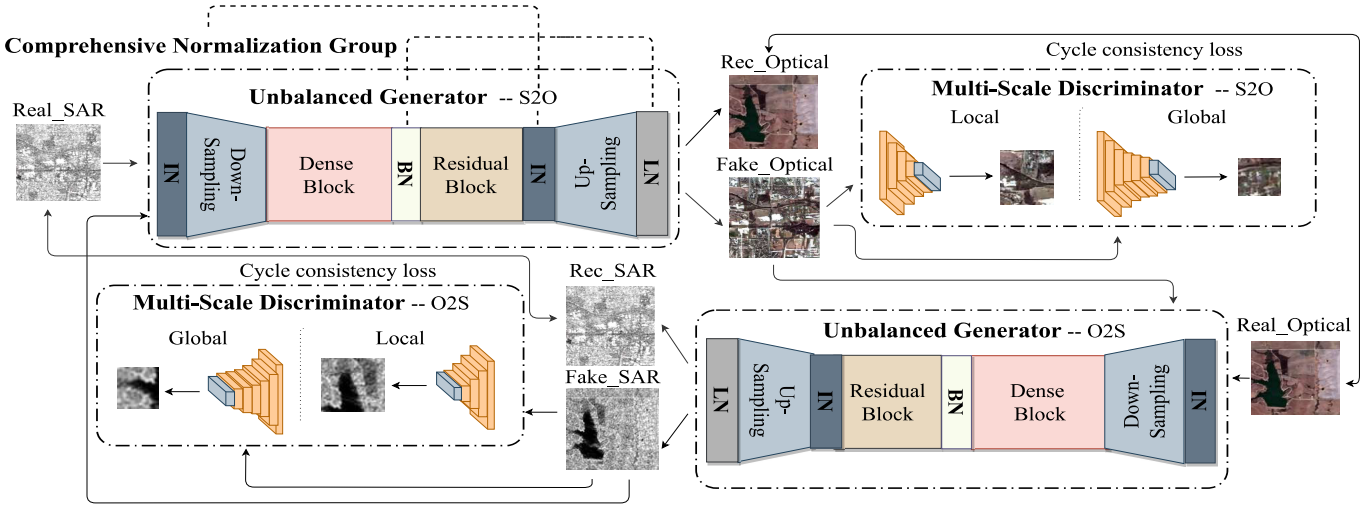


Fig. 1. Framework of FG-GAN. FG-GAN contains three modules: UBG, MSD, and CNG combined with IN/BN/LN.

reusing discriminators to encode images in the target domain, a new unsupervised image-to-image translation framework no-independent-component-for-encoding GAN (NICE-GAN) [21] was proposed. Moreover, Akhtar and Liu [22] presented the cartoonlization style transfer method using multiple discriminators in GAN. In addition to single style transfer, StarGAN [23], MUNIT, and DivCo-DRIT achieved the translation between multiple styles.

B. SAR-to-Optical Image Translation

The emergence of GAN and its derived models provides a new idea for S2O image translation. Most of the existing methods are based on models such as CGAN, Pix2pix, and CycleGAN. In 2018, Schmitt *et al.* [24] released the SEN1-2 dataset, which realized the S2O image translation using Pix2pix and promoted the research of deep learning for SAR and optical data fusion. In view of the limitations in the translation process from S2O image, Fuentes *et al.* [25] proposed some optimization strategies based on CGAN. However, the generated images lacked some details. In 2020, Turnes *et al.* [26] put forward a CGAN network architecture based on atrous convolution. The proposed generator and discriminator combined with atrous spatial pyramid pooling (ASPP) module and used spatial background to enhance the fineness of the generated optical images at multiple scales. Later, to solve complex scenes in optical images and high-frequency speckle noise in SAR images, a feature-guided S2O image translation method [27] was presented to ensure the extraction of multilayer features during image generation process. In addition, Zhang *et al.* [27] designed a loss function based on discrete cosine transform to filter out high-frequency noise to obtain higher quality generated images. Meanwhile, the loss function also affected the quality of the generated images. In 2020, Hwang *et al.* [28] proposed a strongly constrained genetic algorithm based on structural similarity index (SSIM) [29] and L1 norm on the basis of CGAN. The constraint between the generated and target images in the structural information space was established to generate images with better architectural information. Most of these methods are supervised and require paired training samples.

In contrast, there is a lack of efforts spent on using unsupervised methods for S2O image translation. In 2018, Liu and Lei [30] verified the potential of SAR and optical image translation using CycleGAN. Later, Hwang *et al.* [28] used SSIM loss and perceptual loss to maintain rich texture information of generated images. The experimental results showed the superiority of the combination of loss functions. Liu *et al.* [31] realized the change detection of heterogeneous remote sensing images based on CycleGAN. To further improve the quality of the generated image, Baier *et al.* [32] synthesized optical images from land cover maps and auxiliary raster data using GANs. The optical images generated by these methods usually lack accurate color information, which is not conducive to the human eye to correctly recognize the land cover types.

Based on the aforementioned research, an unsupervised S2O image translation method paying more attention to detail and color information is necessary.

III. Proposed Method

We propose an FG-GAN to address the problem of how to enhance the details and alleviate the color distortion of the generated images in S2O image translation tasks. In this work, FG-GAN is a new network architecture proposed based on CycleGAN. FG-GAN has two reciprocal directions of translation, i.e., S2O and optical-to-SAR (O2S). Each direction consists of a UBG module and a multiscale discriminator (MSD) module. We focus on the direction from S2O. The overall architecture of the model is illustrated in Fig. 1. The CNG-applied UBG module receives an SAR image, maps it to the potential space through an encoder, and then remaps it to a translated optical image by a decoder. The generated optical images are input to the local discriminator and global discriminator of MSD for multiscale discrimination, respectively. The purpose of MSD is to identify fake optical images, while UBG is expected to convert SAR images into optical images as realistically as possible, making it unfeasible

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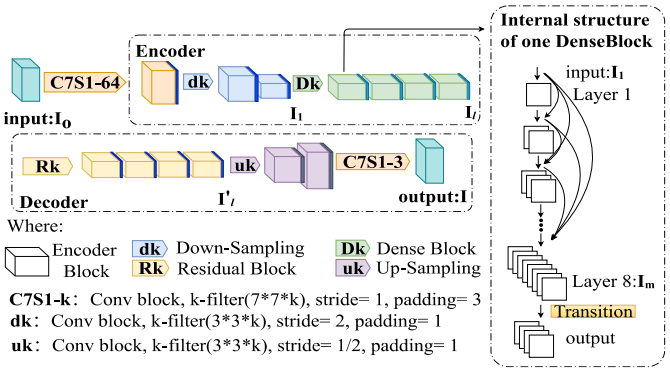


Fig. 2. Network architecture of UBG.

TABLE I
NETWORK ARCHITECTURE OF UGB

Module	Input	Output	Layer Information
Initial Conv	(256*256*3)	(256*256*64)	C7S1-64
Down-sampling	(256*256*64)	(128*128*128)	d-128
	(128*128*128)	(64*64*256)	d-256
DenseBlock	(64*64*256)	(64*64*288)	D32-8-1
	(64*64*288)	(64*64*320)	D32-8-1
	...	...	D32-8-1
	(64*64*480)	(64*64*512)	D32-8-1
	(64*64*512)	(64*64*256)	D32-8-1
ResBlock	(64*64*256)	(64*64*256)	R-256
Up-sampling	(64*64*256)	(128*128*128)	u-128
	(128*128*128)	(256*256*64)	u-256
Conv	(256*256*64)	(256*256*3)	C7S1-3, Tanh

for MSD to separate the two. On the other direction, the network is the same with the only difference being optical as the input and SAR as the translated images. The learning ability of UBG and MSD is improved continuously through adversarial training, which promotes the model to generate images closer to the target domain.

To generate optical images with rich details and color information, we propose three improved methods, including UBG, MSD, and CNG.

A. Unbalanced Generator

Land types represented by SAR and optical images contain a large variety of different texture, boundary contour, and color information. This requires that the network must be able to extract sufficient feature information to generate images relatively accurately. To avoid insufficient feature extraction due to the simplification of networks, we design a complex structure generator which contains dense block and residual block to form an unbalanced network structure.

We add a dense connection mechanism [33] to the encoder, namely, the dense block module in Fig. 1, which fully guarantees the close connection of information flow among all layers. As shown in Fig. 2, given an image $I_o \in \mathbb{R}^{h\times w\times 3}$, we first transform it into $I_1 \in \mathbb{R}^{h/L^2\times w/L^2\times c}$ by down-sampling, where L denotes the number of down-sampling layers, and c represents the number of channels. Then, I_1 is fed into the dense block module. In each dense block, we realize the interconnection between network layers, i.e., each layer accepts the input of all the previous layers and connects these feature maps on the channel dimension. With each dense layer, the number of channels for the feature map increases by k. Then the number of channels for I_m is converted from $(c + km)$ into c via the transition layer to get the final output of one dense block. In the encoder, we apply four dense blocks, each containing eight dense layers and one transition layer. The growth rate k is set to 32, i.e., each layer outputs 32 feature maps after convolution. After that, the feature map $I_l \in \mathbb{R}^{h/L^2\times w/L^2\times c}$ generated from the dense block module are input to residual block of the decoder. Four residual blocks are used to further filter the feature information. Finally, the generated optical image $I \in \mathbb{R}^{h\times w\times 3}$ is obtained via up-sampling the output of residual block $I_l' \in \mathbb{R}^{h/L^2\times w/L^2\times c}$.

Table I shows details of the network architecture of UBG, which is far more complex than ResNet [34]. The unbalanced network structure fully ensures the integrity of feature extraction and improves the restoration of important feature information.

B. Multiscale Discriminator

Given the relatively complex network structure of UBG, a single discriminator cannot perform equal adversarial training with it. Therefore, to match the learning ability of UBG, we present a multiscale discriminant architecture to enhance the discriminant ability of the model, so as to guide the generator to generate images closer to the target domain.

To distinguish real images from the generated images correctly, the discriminator first needs to have a large receptive field to accurately determine the land-type style of the whole image. Second, images with small deviations, such as the lack of complex buildings, contour structure, and other details, can also be judged as fake images by the discriminator. This requires the discriminator to have a small receptive field and focus on some local information. In Fig. 1, MSD contains a local discriminator (D_L) and a global discriminator (D_G). They have similar network structures but discriminate feature maps at different scales. Both D_L and D_G are implemented by PatchGAN [3], i.e., D_L classifies whether the patch of 70×70 generated image is true or false while D_G discriminates between 286×286 patches. Both the discriminators are composed of a down-sampling module and a classifier module. D_L contains three down-sampling layers with smaller receptive fields, thus encouraging the generator to generate more detailed images. D_G applies five down-sampling layers, which enlarges the receptive field. Using the global image perspective, MSD promotes the generation of globally consistent images and avoids the deviation of the overall style of generated images.

Therefore, the learning problem becomes a multitask learning problem of

$$\text{Loss} = \min_{G} \max_{D_L, D_G} [L_{\text{GAN}}(G, D_L) + L_{\text{GAN}}(G, D_G)] \quad (1)$$

where the objective function $L_{GAN}(G, D)$ is given by

$$L_{\text{LSGAN}}(G, D, X, Y) = E_{y\sim P_{\text{data}}(y)}[(D(y) - 1)^2] + E_{x\sim P_{\text{data}}(x)}[D(G(x))^2]. \quad (2)$$

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TABLE II
NETWORK ARCHITECTURE OF MSD

	Local Discriminator			Global Discriminator		
Module	Input	Output	Layer Information	Input	Output	Layer Information
Down-sampling	(256*256*3)	(128*128*64)	C-64	(256*256*3)	(128*128*64)	C-64
	(128*128*64)	(64*64*128)	C-128	(128*128*64)	(64*64*128)	C-128
	(64*64*128)	(32*32*256)	C-256	(64*64*128)	(32*32*256)	C-256
	(32*32*256)	(31*31*512)	C-512	(32*32*256)	(16*16*512)	C-512
				(16*16*512	(8*8*1024)	C-1024
				(8*8*1024)	(7*7*2048)	C-2048
Classifier	(31*31*512)	(30*30*1)	C-1	(7*7*2048)	(6*6*1)	C-1

Here, x is the image from the source domain X and y from the target domain Y.

Therefore, when a real image or a generated image is input to the discriminator, MSD simultaneously discriminates the overall style and local details of the image, which not only improves the capability of the discriminator but also guides UBG to generate optical images with more detailed information while maintaining the overall style of the image. It largely compensates for the lack of details and color distortion in CycleGAN.

Table II shows details of the network architecture of MSD. In the down-sampling module, except the stride of the last layer is 1, $C-k$ is a 4×4 Convolution + InstanceNorm + LeakyReLU layer with k filters and stride 2. The Leaky rectified linear units (ReLUs) are used with a slope of 0.2. In the classifier module, $C-k$ is a 4×4 Convolution layer with k filters and stride 1.

C. Comprehensive Normalization Group

Due to the great difference in imaging mechanism between SAR and optical images, the extracted physical representation information is also significantly different. Therefore, to further improve the similarity between the generated and real images, we use different normalization functions to promote the similarity of physical representation in the process of S2O image translation. CNG includes BN, IN, and LN. BN calculates the mean and variance of each batch along the channel, forcing the data to maintain the positive distribution with a mean of 0 and a variance of 1. It avoids gradient disappearance and provides a solid foundation for the subsequent network. UGATIT pointed out that IN has the effect of removing the style change by directly normalizing the feature statistics of images, and it is more frequently used in style transfer than BN or LN. Since IN only calculates the mean and standard deviation within a channel, it learns the texture features well, but often ignores the learning of overall image style. However, LN is normalized in the channel direction, which affects the image style and color information. Therefore, LN transfers image style features better and focuses on color changes in the process of image translation. In Fig. 1, to combine the advantages of the three, we use BN in dense block to ensure that the network extracts sufficient feature information. Then, IN is applied in residual block to make the UBG better focus on texture, contour, and other details. Since the up-sampling of the decoder generates the target domain image from the embedded features, we use LN in the up-sampling part to promote the generator's attention to color transformation and learning of the overall image style.

Through comprehensive use of UBG, MSD, and CNG, we make up for the defects of the existing unsupervised image translation model with relatively simple structure, improve the overall discrimination ability of the network, and successfully alleviate color distortion and detail missing in the process of unsupervised S2O image translation.

IV. EXPERIMENTAL RESULT AND ANALYSIS

A. Experimental Settings

1) Baseline: We use CycleGAN as baseline and compare FG-GAN with the SOTA methods in recent years including MUNIT, UGATIT, and DivCo-DRIT. To ensure the fairness of the experiment, we use the same number of training times and the official public code.

2) Datasets: The SEN12 [24] dataset comprises 282 384 SAR optical patch pairs acquired by Sentinel-1 and Sentinel-2. The patches are collected from locations spread across the land masses of the Earth and over all four seasons. Sentinel-1 data contain VV-polarized images. The images from Sentinel-2 have three bands: red, green, and blue.

The SEN12MS [35] dataset contains 180 662 patch triplets of the corresponding Sentinel-1 dual-pol SAR data, Sentinel-2 multispectral images, and MODIS-derived land cover maps. The patches are distributed across the land masses of the Earth and spread over all four meteorological seasons. Sentinel-1 data contain both VV- and VH-polarized images. We extracted the VV directional polarization channel of the image and used it as the input image (SAR) of the model. In addition, Sentinel-2 multispectral data contain 13 channels, and only three channels (RGB) were used to synthesize the required optical images.

The WHU-SEN-City [36] dataset which covers 32 Chinese cities contains 18 542 paired training samples and 4566 paired test samples. Sentinel-1 SAR data have two bands, which are VH and VV with a spatial resolution of 20 m in the range direction and 22 m in the azimuth direction. The authors used VH and VV bands to generate a new band VH/VV. Therefore, the input SAR data have three bands. As with SEN12 and SEN12MS, only red, green, and blue bands of the Sentinel-2 images were used as optical.

We selected the training and test images from ten scenes of the SEN1-2 dataset. Considering the influence of training time and GPU memory on the experimental efficiency, we randomly selected 3000 SAR images and 3000 optical images from the SEN1-2 dataset for unpaired training and 800 SAR images and

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TABLE III
NUMBER OF SCENE DATA PER CATEGORY OF THE SEN12 DATASET

Category	Train	Test
Mountain	293	49
Urban	312	85
Forest(spring)	350	95
Forest(summer)	286	79
Semi-urban	484	95
Rural	121	22
Farmland	368	110
Plain	251	80
Grassland	247	78
Others	288	107

800 optical images for testing. For each scene, the training set and test set were selected uniformly and noncross. Moreover, the distribution of training data and test data is similar but nonoverlapping. In addition, we divide these training and test images into urban, semiurban, rural, grassland, forest, and other scenes as shown in Table III. Since 3000 images from the SEN12 dataset are selected, the dataset was augmented to avoid the phenomenon of overfitting in the training process. We first adjusted the size of the image from 256 $\times$ 256 to 286 $\times$ 286 by bicubic interpolation, then randomly flipped it left and right, and finally randomly cropped the image to the same size of 256 $\times$ 256 as the final input image. The selection process of the SEN12MS dataset was the same as that of SEN12, but considering that the SEN12MS dataset contains abundant land types, to prevent the overfitting phenomenon, 6000 images were selected as the training set, and the test set was still 800. For the WHU-SEN-City dataset, we selected a total of 6000 images from eight cities as the training set and 800 images from two cities as the test set.

3) Setup: We do experiments on the computer with a single GTX TITAN with 12-GB GPU memory and implement on PyTorch. In all, 200 epochs are trained on the SEN12 dataset and 100 for SEN2MS and WHU-SEN-City. We use Adam optimizer with the batch size of 1, and the exponential decay rates of control redistribution and gradient square are set to 0.5 and 0.999, respectively. To stabilize model training, the least-square loss function proposed in LSGAN is used to punish the generated image distribution which is far away from real image data distribution and make it closer to real image distribution. Meanwhile, in the training process, identity loss is added to better constrain the similarity between the generated image and the real image. Among them, the parameter for cycle consistency loss and identity loss is, respectively, set to 10 and 5 for balancing the generated result of FG-GAN.

4) Evaluation Metrics: We conduct evaluations using both subjective and objective evaluation indicators. Peak signal-to-noise ratio (PSNR), SSIM, and Frechet inception distance (FID) [37] are applied as subjective evaluation indicators, and natural image quality evaluator (NIQE) [38] and learned perceptual image patch similarity (LPIPS) [39] as objective evaluation indicators. **PSNR** and **SSIM** are often used for image quality evaluation. Generally, higher PSNR values and SSIM values represent the better quality of the generated image. **FID** is applied to measure the distance between the generated image distribution and the real image distribution through features extracted from the initial network. **NIQE** uses the distance between the image feature model parameters to be evaluated and the preestablished model parameters to determine the image quality. **LPIPS** measures the average feature distances between the generated samples. Lower values of FID, NIQE, and LPIPS indicate the smaller difference between the generated image and the real image.

B. Results of SAR-to-Optical Image Translation

We apply three methods to verify the similarity between the generated image and the real image. First, we compare the visualization results of the generated images with the real images. Furthermore, five evaluation indicators including PSNR, SSIM, FID, NIQE, and LPIPS are used to evaluate our model. Therefore, we can test generated images from both subjective and objective evaluation indicators. Finally, we conduct classification experiments using the optical images generated by FG-GAN and other SOTA methods.

1) Visualization Comparison: Figs. 3–5 show the optical images generated by different unsupervised methods on three datasets. By comparison, we find that MUNIT has insufficient learning ability for contour boundary and texture information, resulting in the generated optical images with significant missing details. UGATIT and CycleGAN basically simulate the contour structure of optical images and describe the boundary information of the region, but the details of the generated image are not fine enough. DivCo-DRIT could learn some contour and texture information, but it translates the input images of different land types into the same type of optical images on the SEN12 dataset as shown in Fig. 3, which seriously affects the recognition of land types.

In addition to the lack of detail learning ability, we also find that the generated images of these four models have obvious color distortion, i.e., the land types of the generated images are significantly different from the land types displayed by the real optical images. For example, in the first and third rows of Fig. 3, the land types of the real optical images are urban construction areas, but the four models generate images of land types representing forest or grassland. In Fig. 4, the image generated by CycleGAN has noticeable color deviations from the real image on the SEN12MS dataset. These phenomena have seriously affected the correct evaluation of land types represented by SAR images.

In contrast, FG-GAN proposed in this article alleviates these phenomena. It helps fully extract the feature information; so that some details and color information are well-learned. For example, from the first three rows in Fig. 3, it can be seen that the optical image generated by FG-GAN not only contains rich details and the distribution of white buildings but also maintains the correct land type. For semiurban or rural areas with fewer buildings (fourth and sixth rows in Fig. 3), FG-GAN performs better than the other four models, basically learning the distribution of buildings and color information of grassland. As shown in Figs. 4 and 5, FG-GAN generates images with richer texture detail and relatively accurate color features compared with SOTA methods.

2) Metrics Comparison: Tables IV–VI show the evaluation results of FG-GAN and other SOTA methods on the SEN12,

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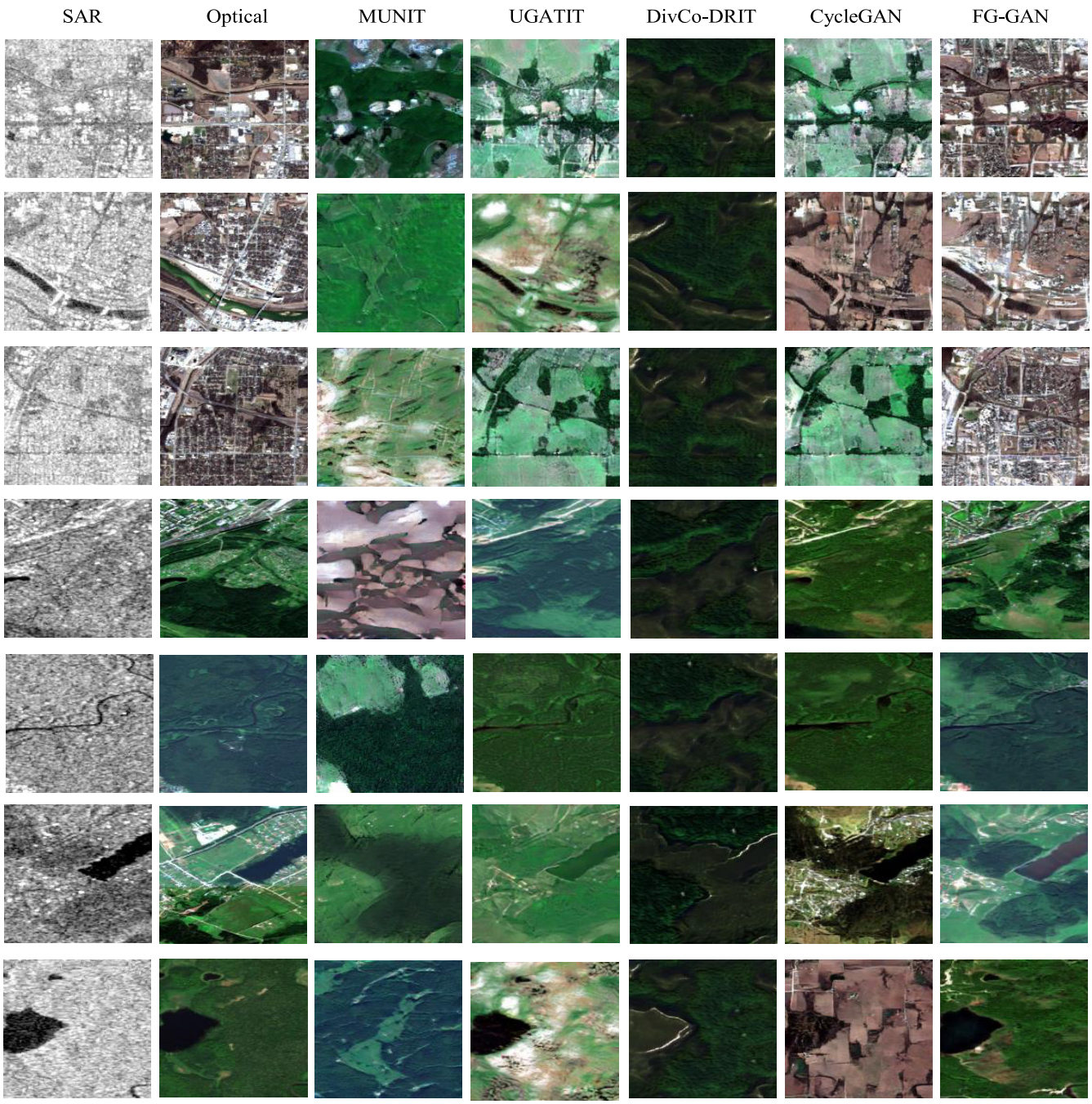


Fig. 3. Visualization results of different models for S2O image translation on the SEN12 dataset.

TABLE IV
EVALUATION INDEX RESULT ON THE SEN12 DATASET (S2O)

Method	PSNR ↑	SSIM ↑	NIQE ↓	FID ↓	LPIPS ↓
MUNIT	10.81	0.1325	21.2	122.6	0.669
UGATIT	10.71	0.1312	23.4	110.6	0.661
DivCo-DRIT	11.05	**0.1931**	**19.4**	198.8	0.665
CycleGAN	11.18	0.1402	21.4	101.8	0.588
FG-GAN	**11.34**	0.1450	20.9	**97.3**	**0.569**

SEN12MS, and WHU-SEN-City datasets. respectively. Bold indicates optimal and blue means suboptimal. Our method gets good scores on the majority of evaluation indicators, and most of the results are in the top two. Especially in the PSNR and LPIPS evaluation indicators, our method achieves the best results on three datasets.

3) Classification Comparison: Data scene classification is to further verify that the optical images generated by the FG-GAN model proposed in this article are closer to the real optical images. We construct a single ten classification model using Visual Geometry Group (VGG) [40] network with real optical images as the training set and optical images generated by different models as the test set.

In the classification experiment of the SEN12 dataset, combined with Fig. 3, it can be seen that the images generated

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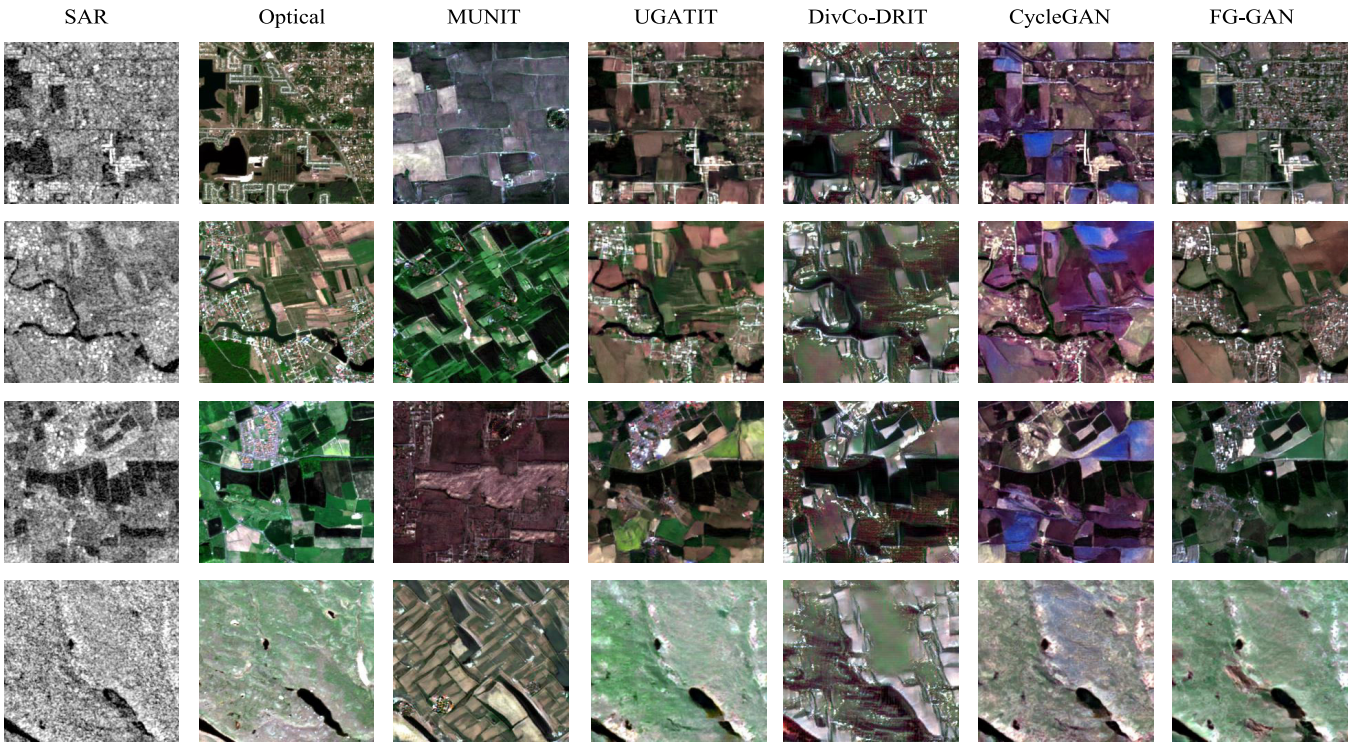


Fig. 4. Visualization results of different models for S2O image translation on the SEN12MS dataset.

TABLE V
EVALUATION INDEX RESULT ON THE SEN12MS DATASET (S2O)

Method	PSNR ↑	SSIM ↑	NIQE ↓	FID ↓	LPIPS ↓
MUNIT	9.27	0.0727	25.3	105.5	0.612
UGATIT	10.94	0.1220	25.1	106.6	0.613
DivCo-DRIT	10.84	0.1183	**24.0**	183.5	0.605
CycleGAN	10.82	0.1228	24.2	150.8	0.609
FG-GAN	**11.22**	**0.1311**	25.9	**103.1**	**0.602**

TABLE VI
EVALUATION INDEX RESULT ON THE WHU-SEN-CITY DATASET (S2O)

Method	PSNR ↑	SSIM ↑	NIQE ↓	FID ↓	LPIPS ↓
MUNIT	10.09	0.0772	21.3	117.3	0.619
UGATIT	9.73	0.0526	23.8	**75.4**	0.577
DivCo-DRIT	8.31	0.0402	**16.9**	207.9	0.587
CycleGAN	10.83	**0.0912**	21.9	88.0	0.574
FG-GAN	**10.93**	0.0862	21.5	86.8	**0.571**

by FG-GAN in different scenes are closer to real optical images. Therefore, classification accuracy is also higher than other SOTA methods, as shown in Table VII. We also test the accuracy of a single classification on the SEN12 dataset to ensure that our proposed method could precisely learn the characteristics of different scenes. The results of single classification show that FG-GAN can better restore the scenes depicted by SAR images and generate optical images that are closer to real images.

Meanwhile, the similar overall style of SAR interferes with the extraction of style features by the style encoder in DivCo-DRIT, which may generate a single style of optical images, as in column 5 of Fig. 3. It can also be concluded from Table VII that the classification accuracy of DivCo-DRIT is almost zero except for the mountain file which indicates that DivCo-DRIT learns little about the characteristics of other regions. For images in the urban file, the classification accuracy of MUNIT, UGATIT, DivCo-DRIT, and CycleGAN is very low, while FG-GAN gets up to 0.812. However, among the ten types of images, the classification result of FG-GAN for images from the forest (summer) file is the worst, only 0.291. The other four unsupervised models also have relatively low classification accuracy for such images. Therefore, what causes this phenomenon? We find that the images in the urban file contain a large number of detailed building areas with relatively complex texture and contour information. Therefore, such images have high requirements for the detail learning ability of models. However, images in forest (summer) files have minor texture variations, contain less detailed information, and have relatively simple contour structures. It is not conducive to model extraction information, and feature information extracted by the model is comparatively single, resulting in ineffective image translation. Therefore, combining these two points, it can be concluded that the FG-GAN model extracts a large amount of feature information, pays more attention to the learning of details, and performs a good translation of images with rich details to generate finer optical images.

In the classification experiment of the SEN12MS dataset, we first carried out a pixel-level classification experiment according to the land cover label map provided by SEN12MS dataset, and the results are shown in the second line of Table VIII. Due to the deviation between the marking of the land cover map and the categories presented by real

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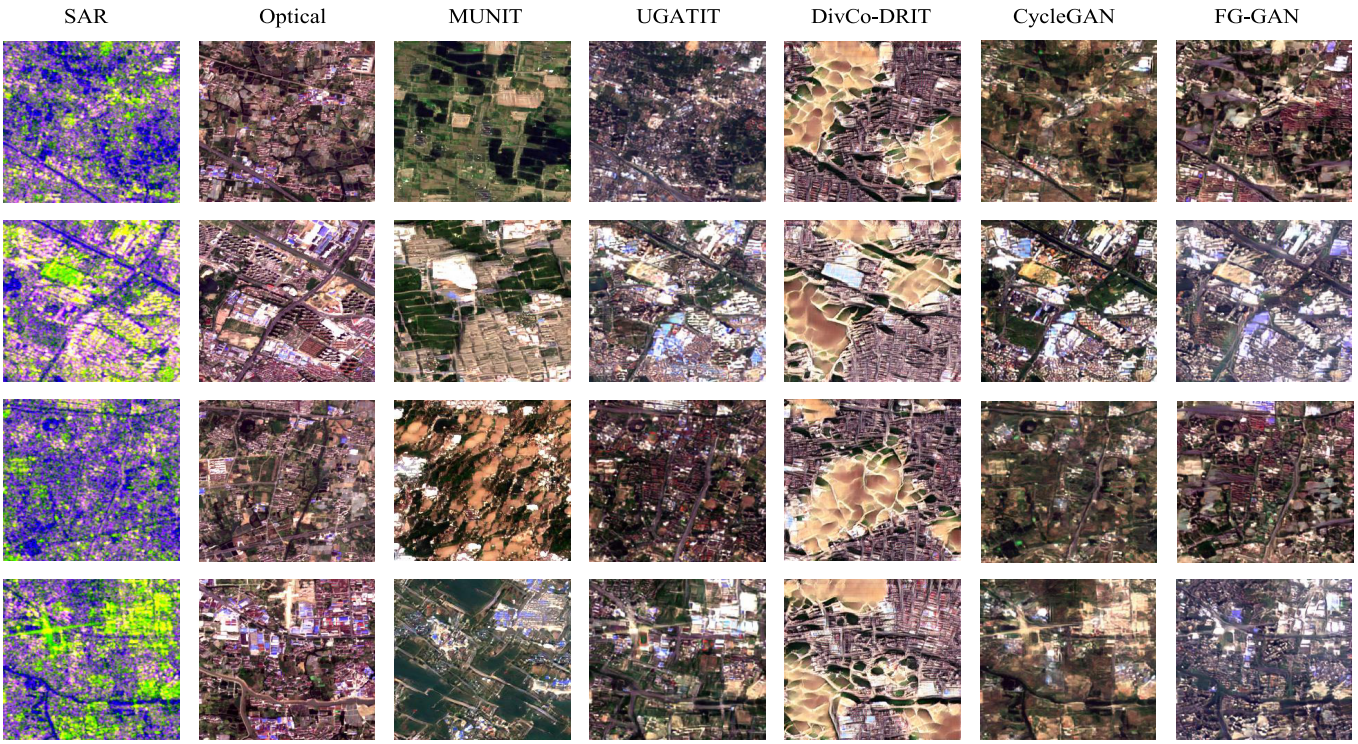


Fig. 5. Visualization results of different models for S2O image translation on the WHU-SEN-City dataset.

TABLE VII
CLASSIFICATION ACCURACY USING DIFFERENT MODELS ON THE SEN12 DATASET

Category	MUNIT	UGATIT	DivCo-DRIT	CycleGAN	FG-GAN
Mountain	0.163	0.489	0.621	0.694	**0.795**
Urban	0.035	0.024	0	0.200	**0.812**
Forest(spring)	0.147	0.578	0	0.650	**0.660**
Forest(summer)	0.063	0.174	0	0.215	**0.291**
Semi-urban	0.176	0.518	0.061	0.452	**0.525**
Rural	0	0.350	0	0.130	**0.381**
Farmland	0.082	0.700	0.018	0.742	**0.782**
Plain	0.150	0.221	0	0.375	**0.500**
Grassland	0.141	0.051	0.013	0.179	**0.610**
Others	0.178	0.439	0	0.682	**0.850**
Average accuracy	0.113	0.354	0.071	0.448	**0.695**

TABLE VIII
CLASSIFICATION ACCURACY BY DIFFERENT MODELS USING DIFFERENT METHODS ON THE SEN12MS DATASET

Mothed	MUNIT	UGATIT	DivCo-DRIT	CycleGAN	FG-GAN
Pixel-level	0.453	0.528	0.525	0.526	**0.533**
Scene-level	0.125	0.279	0.236	0.313	**0.405**

optical images, we adopted the same scene-level classification experiment as SEN12 dataset, and the results are shown in the third line of Table VIII. The classification accuracy of FG-GAN is significantly higher than that of other SOTA methods. Meanwhile, combined with Fig. 4, we conclude that the image generated by FG-GAN is closer to the real optical image. However, due to a large amount of data, complex scenes, and various colors, the SEN12MS dataset has lower classification accuracy than the SEN12 dataset.

By observing the visualization results and classification accuracy, it can be concluded that our FG-GAN model with UBG, MSD, and CNG modules pays more attention to detail learning and performs relatively well in color information representation. Compared with the other four unsupervised image-to-image translation models, the experimental results show the superiority of our method.

TABLE IX
DIFFERENT COMPONENTS IN THE PROPOSED METHOD ON THE SEN12 DATASET

Components	PSNR ↑	SSIM ↑	NIQE ↓	FID ↓	LPIPS ↓
B	11.18	0.1402	21.4	101.8	0.588
B+UBG (w/o CNG)	11.67	0.1395	22.8	100.4	0.610
B+UBG (w CNG)	11.69	0.1405	21.9	99.8	0.614
B+MSD	11.61	0.1425	24.4	124.4	0.650
ALL	**13.34**	**0.1450**	**20.9**	**97.3**	**0.569**

C. Ablation Study

We evaluate each component in the proposed method, i.e., UBG, MSD, and CNG. All the experiments are conducted on the SEN12 dataset, and the results are given in Table IX.

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Compared with the baseline model (B), some of the evaluation indicators with the UBG (w/o CNG) method become better like PSNR and FID, but SSIM, NIQE, and LPIPS get worse. It verifies that the generation capability of UBG is improved. The fake optical image generated by UBG is more likely to fool the discriminator. After introducing CNG, SSIM score is improved from 0.1395 to 0.1405, which is higher than B. It proves that CNG could be further enhanced the consistency between the generated and optical images. But if we only use MSD, all the evaluation indicators have been destroyed. It indicates that the discriminator is more capable and easier to recognize for the generated image. Therefore, it is not suitable for generators with relatively simple structures like B. Finally, our method achieves the best evaluation indicators with all the components on the SEN12 dataset.

V. Conclusion

In this article, a new unsupervised S2O image translation method, FG-GAN, which can generate fine-grained optical images is proposed as an auxiliary tool to solve the visual observation problem of SAR images. Our method uses cycle consistency loss to realize unsupervised S2O image translation, which includes UBG with unbalanced structure, MSD with multiscale and high discrimination ability, and CNG module with multiple normalization functions. Among them, UBG helps extract abundant feature information. MSD is used to accurately discriminate the generated images. CNG promotes the consistency of physical representation of SAR and optical images. In addition, we apply three different ways to verify the similarity between the FG-GAN-generated image and real image, namely, visualization results, five evaluation indicators, and data scene classification and compare FG-GAN with the SOTA unsupervised image-to-image translation methods on three public datasets. The experimental results show the superiority of the proposed method, i.e., FG-GAN, can generate more detailed optical images. Overall, the proposed method opens up a new possibility for the future development of S2O image translation using unsupervised methods, which is beneficial to enhance the interpretability of SAR images.

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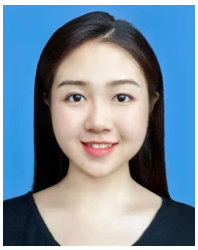
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